Customer Behavior Analysis

Python · SQL · Power BI

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Dataset Overview

3.9K

Customer Transactions

Complete purchase records analyzed

18

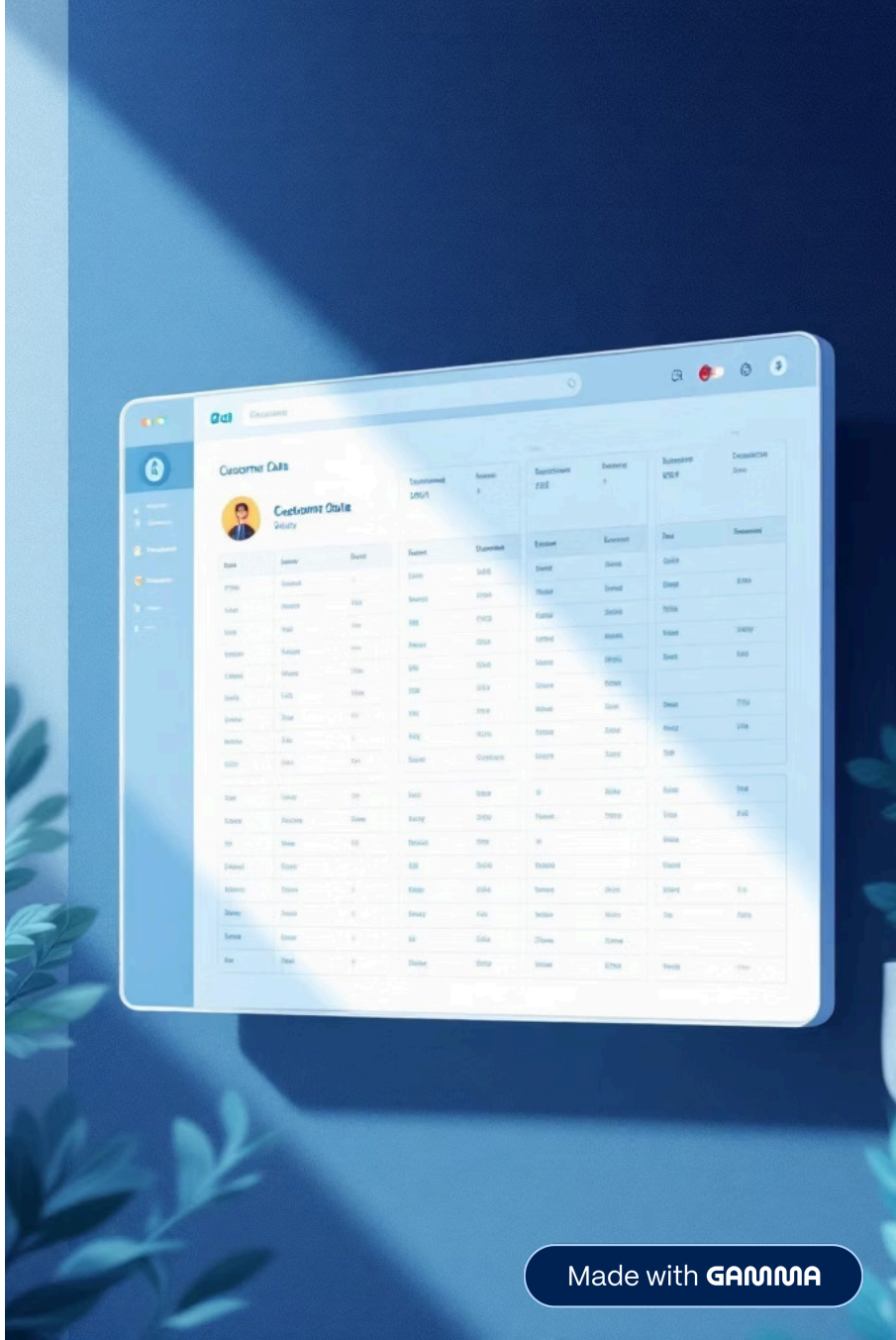
Data Attributes

Demographics, purchases, behavior

3

Product Categories

Clothing, Accessories, Footwear



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Python: Data Foundation

Data Loading & Exploration

- Imported with pandas
- Initial structure analysis
- Statistical summaries

Cleaning & Preparation

- Handled missing values
- Standardized formatting
- Exported to database

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SQL: Revenue Insights

Gender Revenue Split

Male: $157,890

Female: $75,191

High-Spend Discount Users

Identified customers using discounts while spending above average

Top Products by Rating

Gloves (3.86), Sandals (3.84), Boots (3.82)

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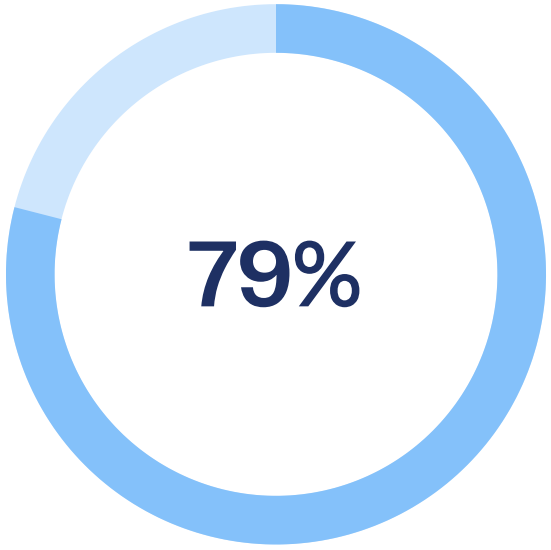
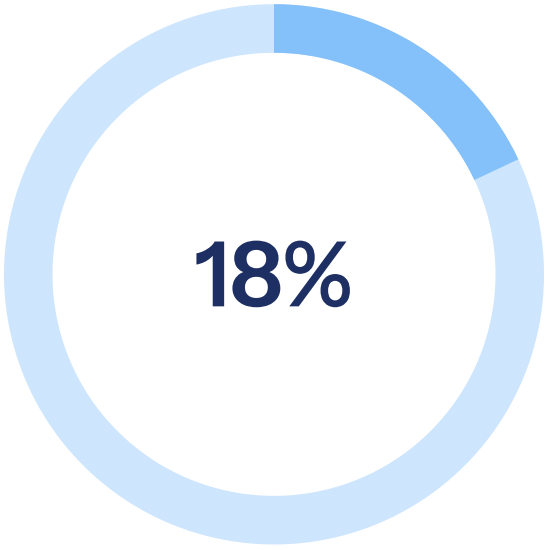
SQL: Customer Segmentation



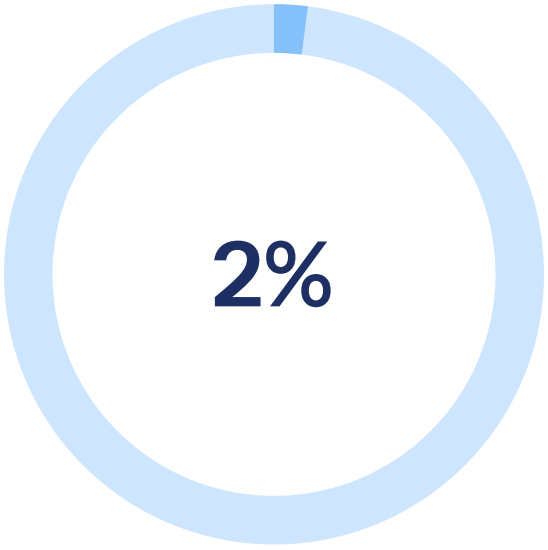
Loyal Customers

3,116 customers with strong purchase history



Returning

701 customers making repeat purchases



New

83 customers with single transactions

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SQL: Key Findings

Shipping Type Impact

Express: $60.48 avg

Standard: $58.46 avg

Discount-Dependent Products

Hat (50%), Sneakers (49.66%), Coat (49.07%)

Repeat Buyers & Subscriptions

958 subscribers vs 2,518 non-subscribers with >5 purchases

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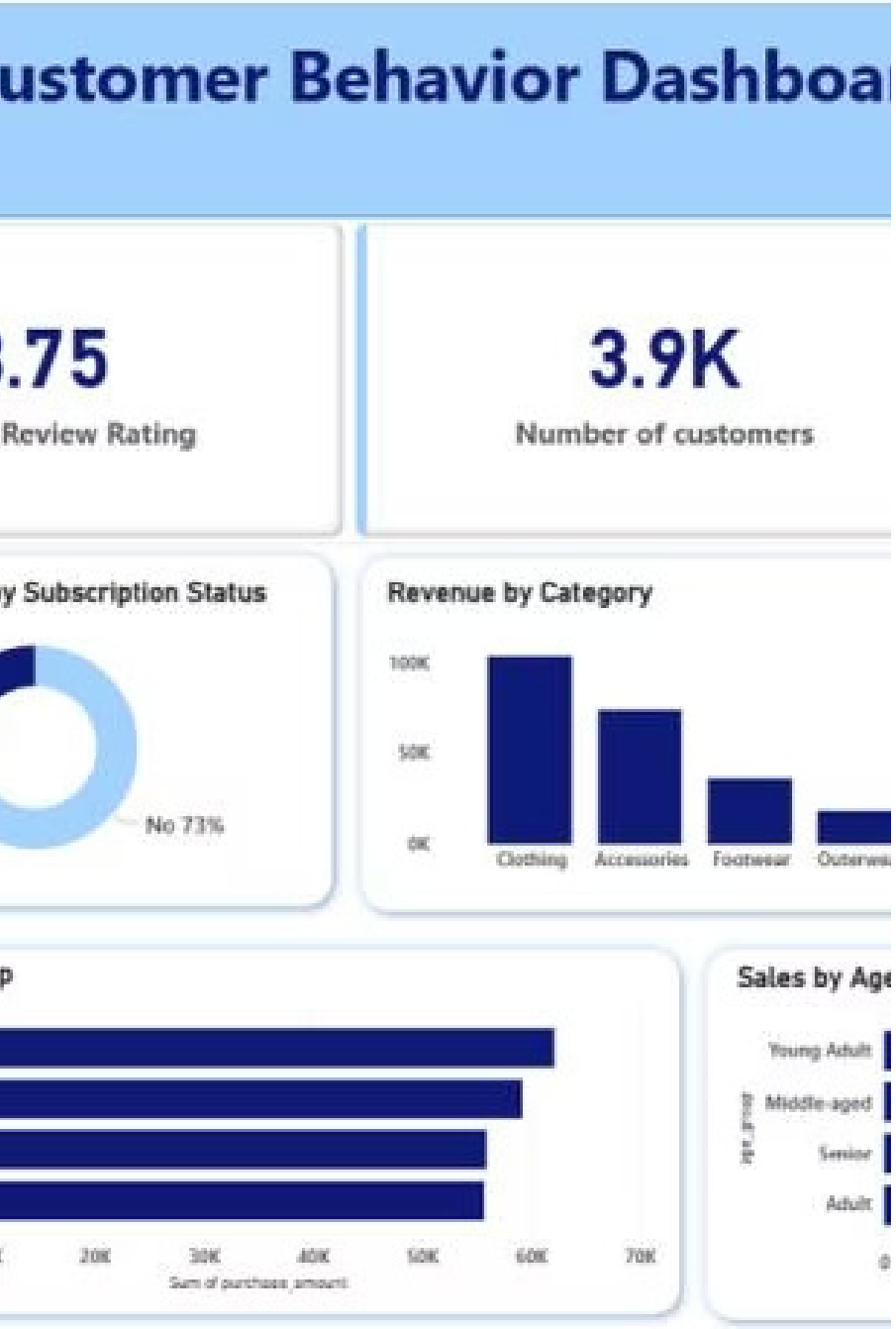

Power BI Dashboard

3.75

Average Review Rating

$59.76

Average Purchase

27%

Subscribers

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Dashboard Highlights

Revenue by Category

Bar chart showing Clothing, Accessories, Footwear, Outerwear performance

Age Group Analysis

Horizontal bars revealing revenue and sales patterns across demographics

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Business Recommendations

01

Boost Subscriptions

Promote exclusive benefits for subscribers

02

Customer Loyalty Programs

Reward repeat buyers to move them into "Loyal" segment

03

Review Discount Policy

Balance sales boosts with margin control

04

Product Positioning

Highlight top-rated and best-selling products in campaigns

05

Targeted Marketing

Focus on high-revenue age groups and express-shipping users

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Key Takeaways

High-Value Segments

Loyal customers drive 79% of revenue

Data-Driven Decisions

Python, SQL, Power BI pipeline delivers actionable insights

Revenue Opportunities

Target male customers and express-shipping users

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